



The cosmic microwave background

Our probe of the birth of the universe

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Content: What are Dark Matter and Dark Energy?

The Cosmic Microwave Background (CMB)

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Reality: Our direct image of the early universe

The Night Sky

The Milky Way





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Ground based telescopes

The Coma Cluster



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Space Telescopes

Hubble Extreme Deep Field



Space Telescopes

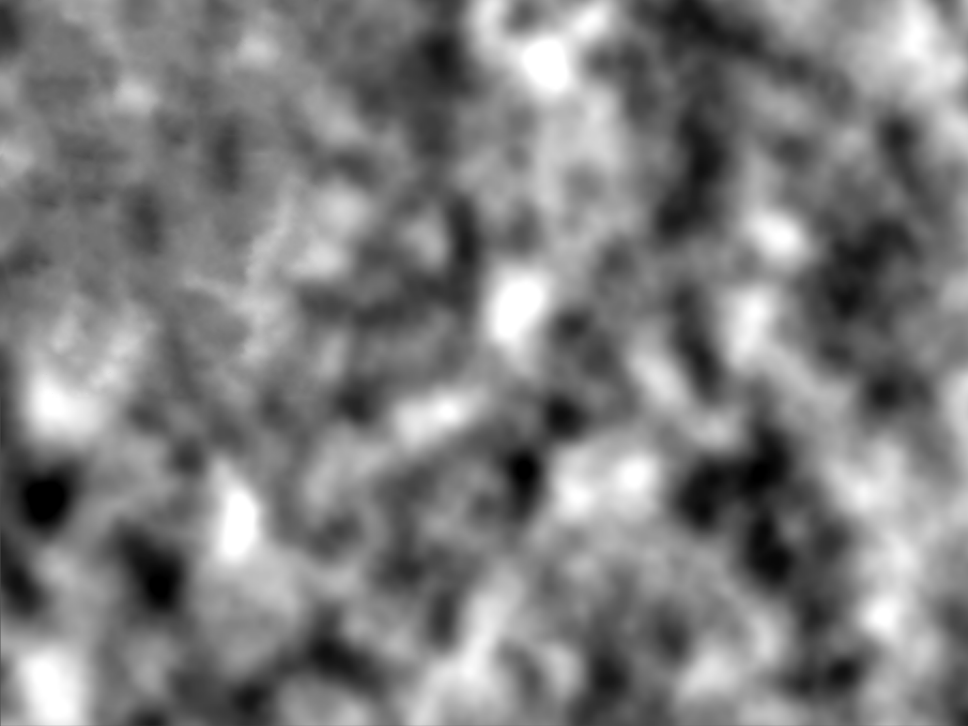
Hubble Extreme Deep Field

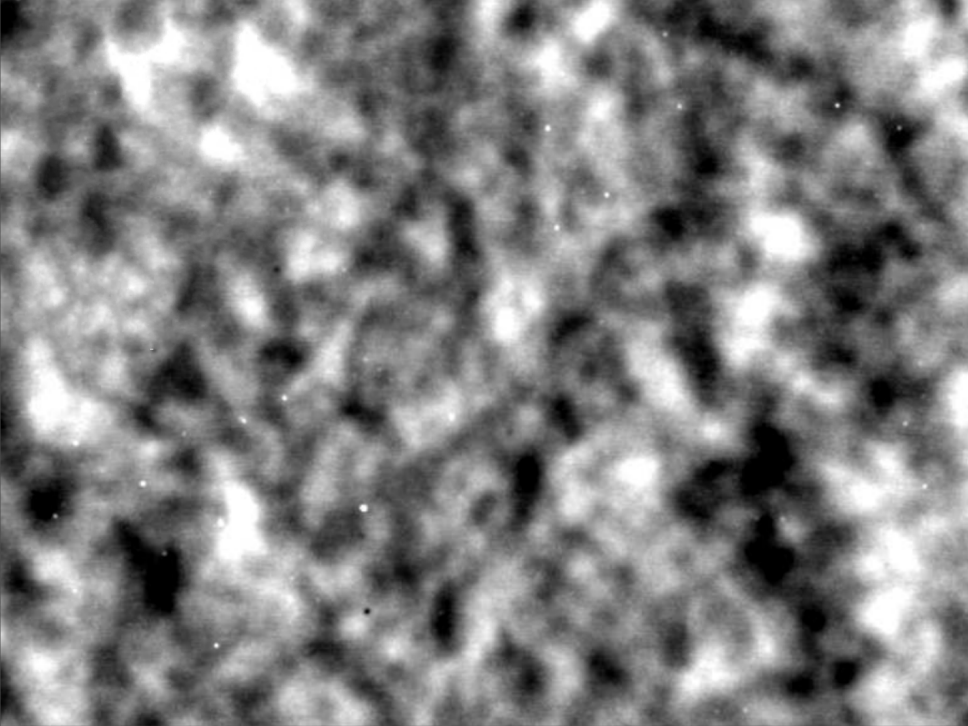


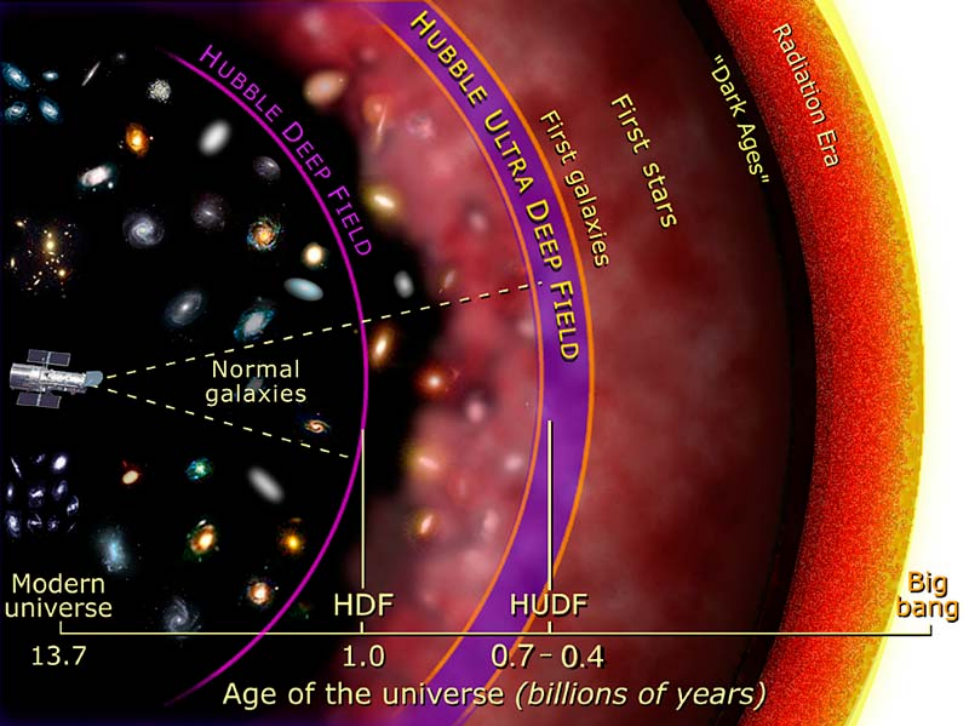
The Dark Ages

The CMB

Cosmic Microwave Background

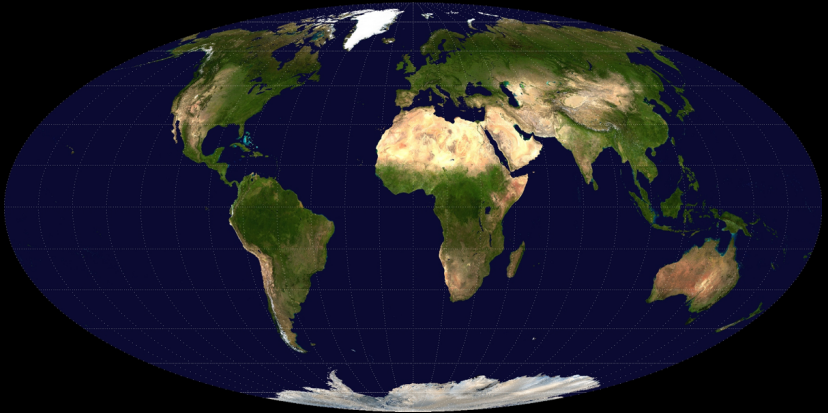






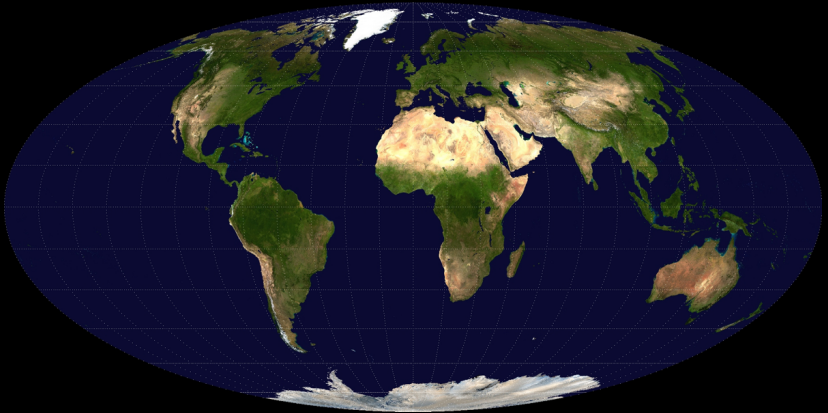
Anisotropies in the CMB

A more interesting picture



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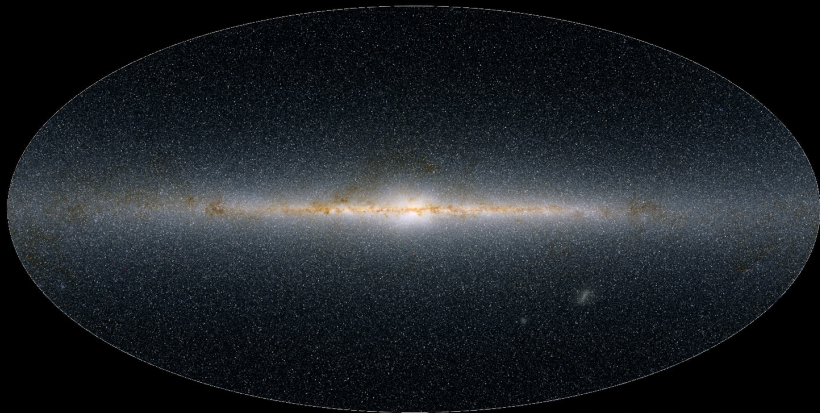
A more interesting picture



► Map of the earth

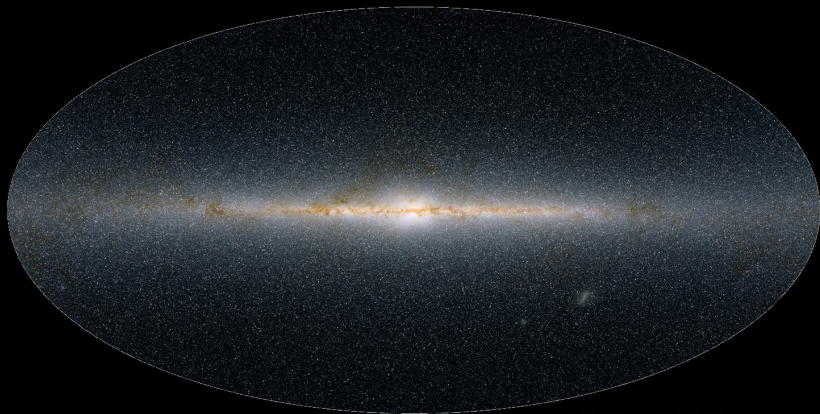
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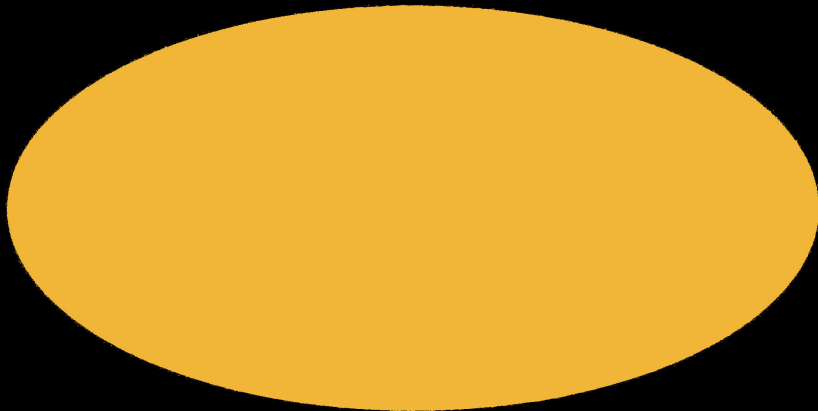
A more interesting picture



► Map of the sky

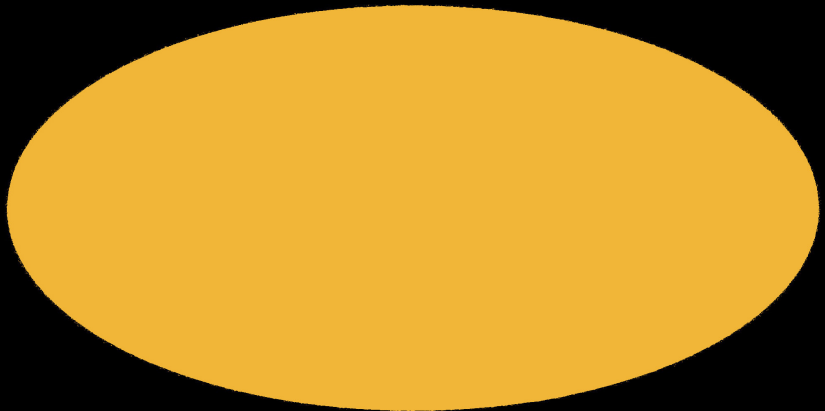
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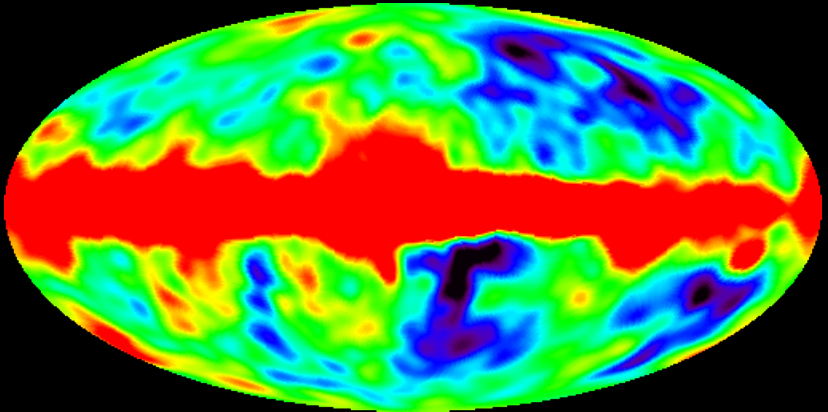
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- ▶ 'Subtract off' the constant source.

Anisotropies in the CMB

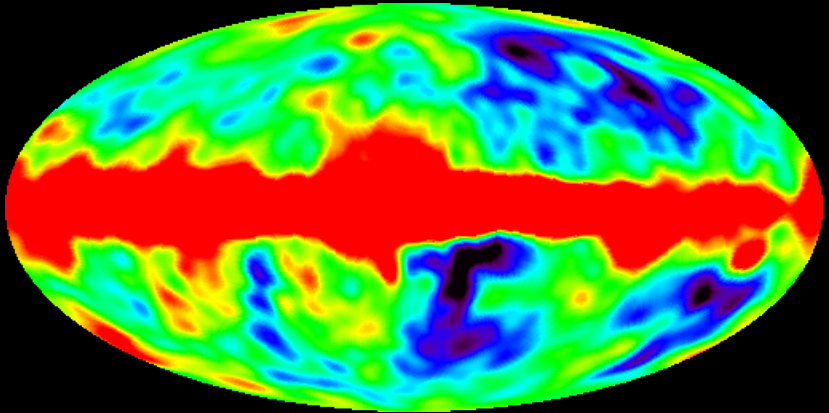
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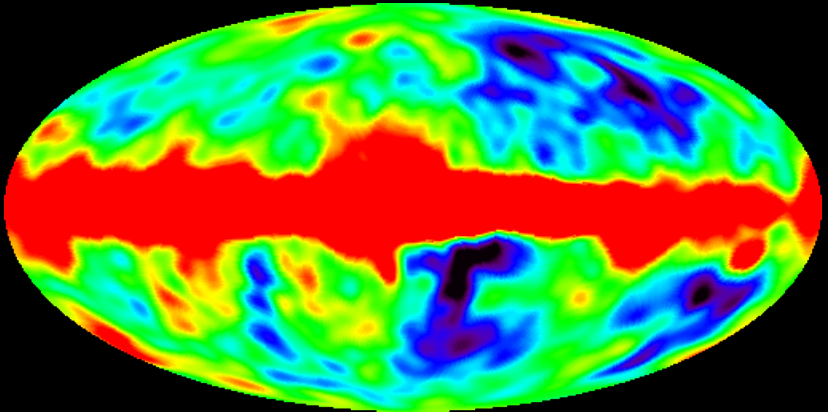
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- ▶ 'Subtract off' the constant source.
- ▶ COBE microwave satellite (1992)

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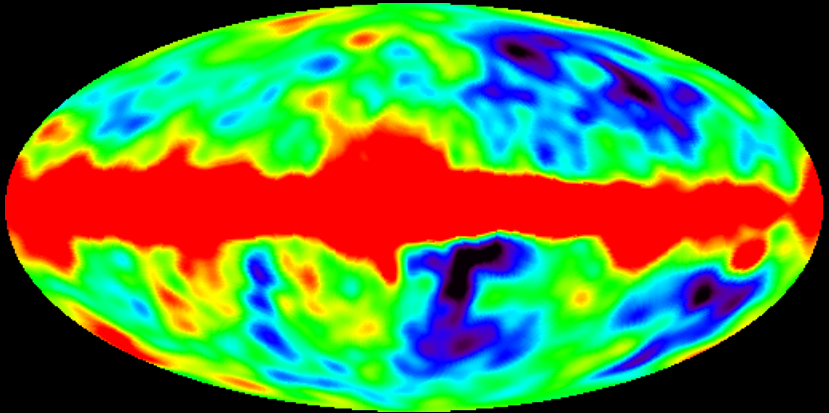
A more interesting picture



- ▶ 'Subtract off' the constant source.
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- ▶ These direction-dependent deviations are called *Anisotropies*

Anisotropies in the CMB

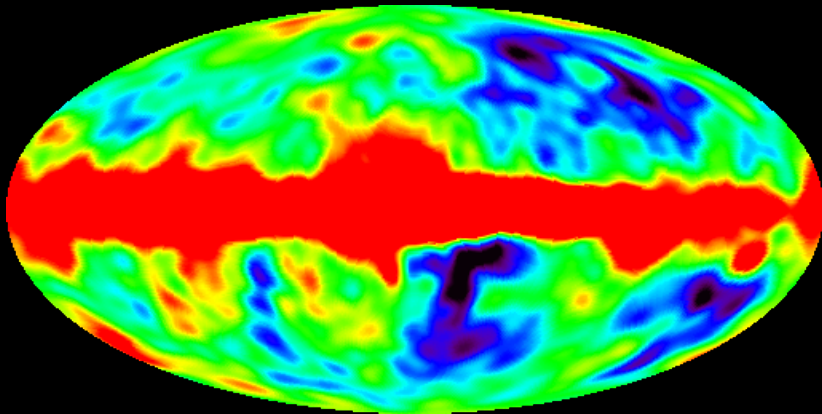
A more interesting picture



1. Galaxy in the middle

Anisotropies in the CMB

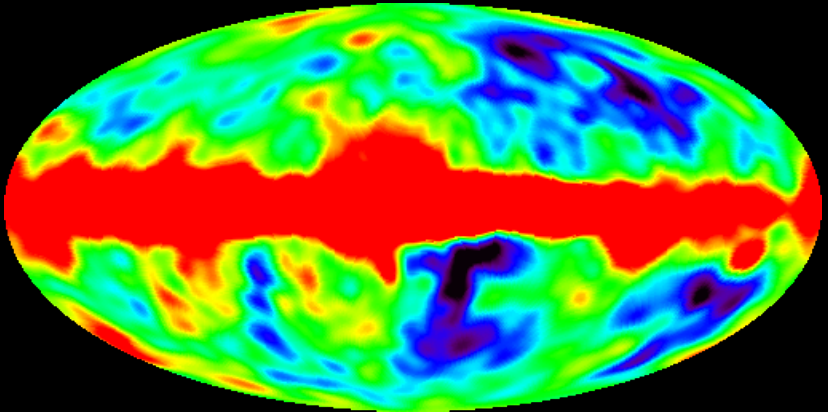
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1. Galaxy in the middle
2. One half hotter than the other

Anisotropies in the CMB

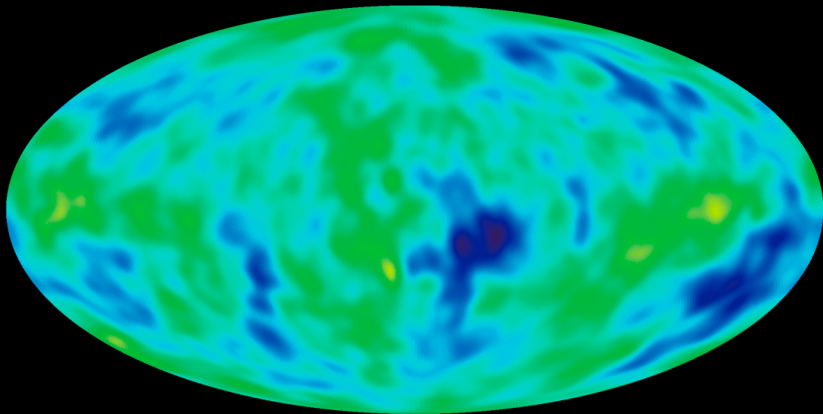
A more interesting picture



1. Galaxy in the middle
2. One half hotter than the other
3. Additional fluctuations

Anisotropies in the CMB

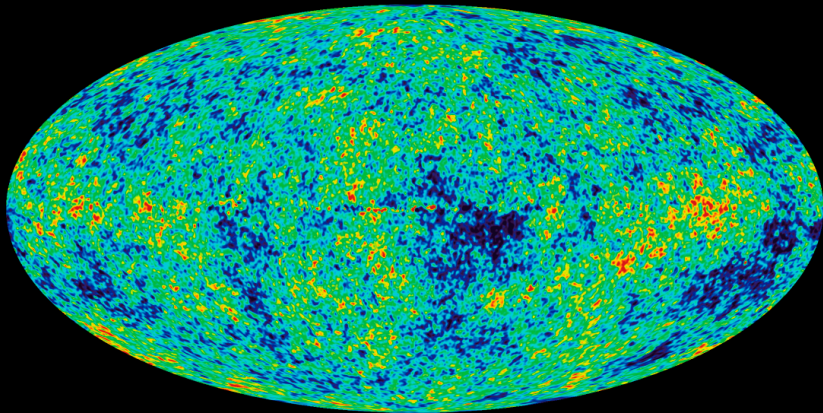
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► COBE (1992)

Anisotropies in the CMB

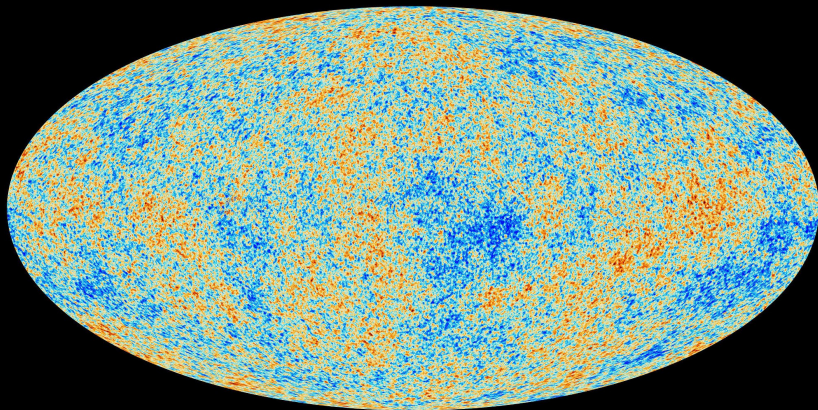
A more interesting picture



► WMAP (2001–2010)

Anisotropies in the CMB

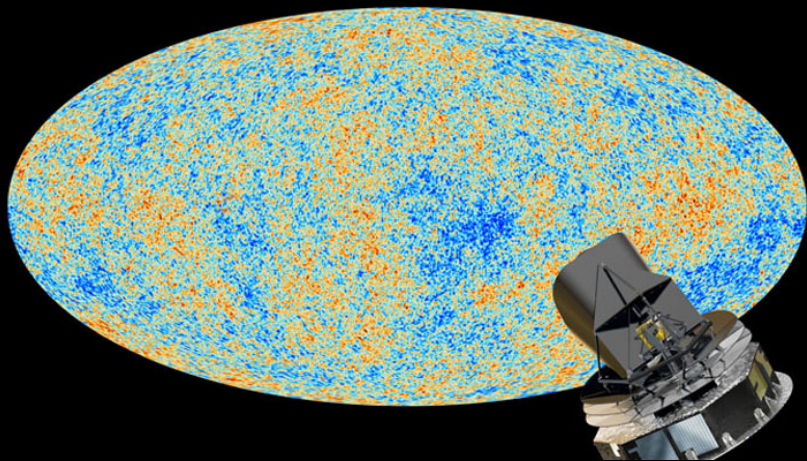
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► Planck (2009–2013)

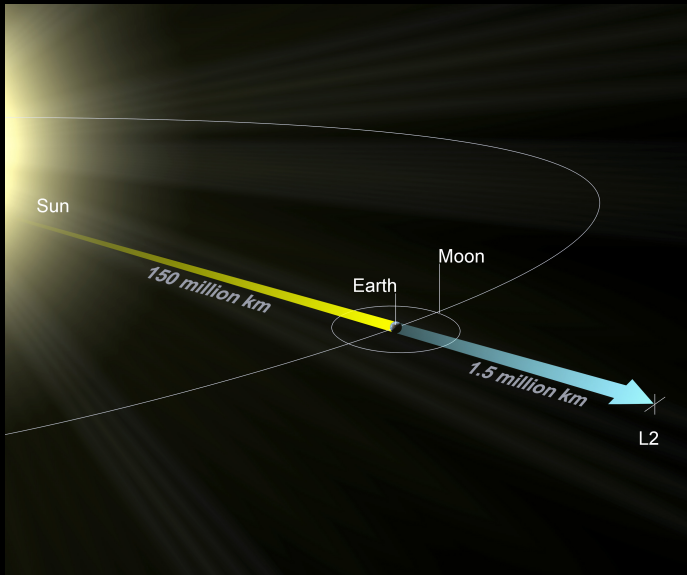
How do we study the CMB?

The Planck satellite

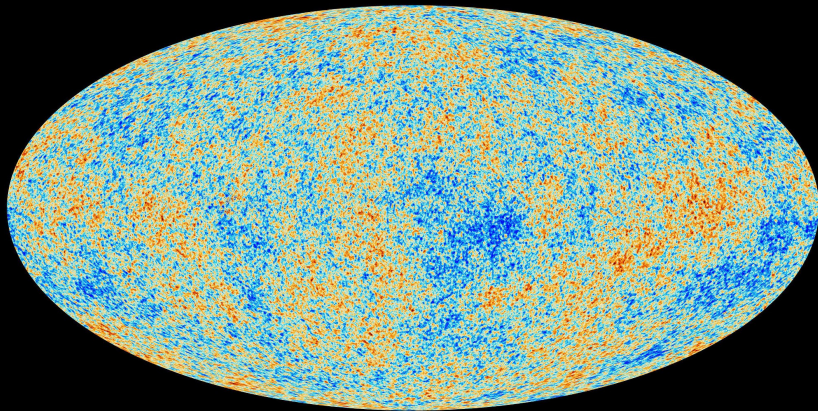


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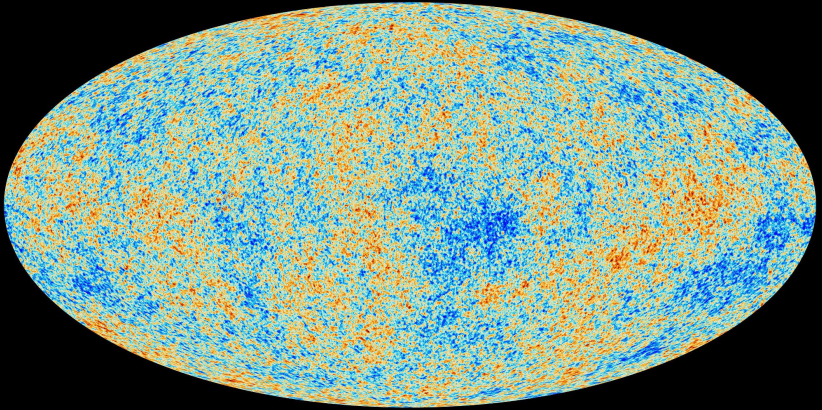
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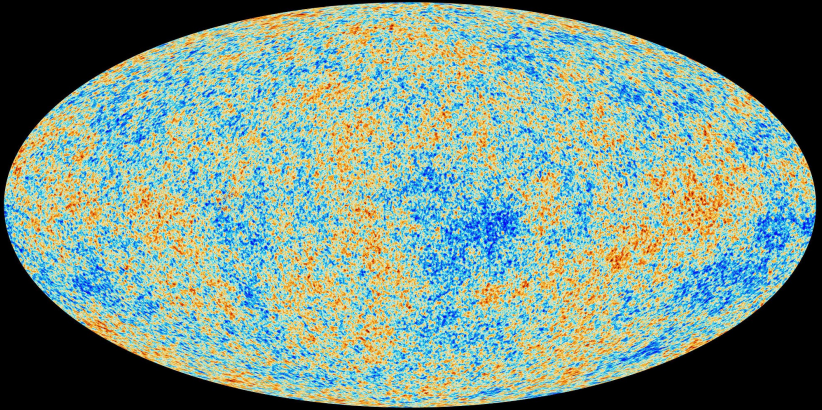


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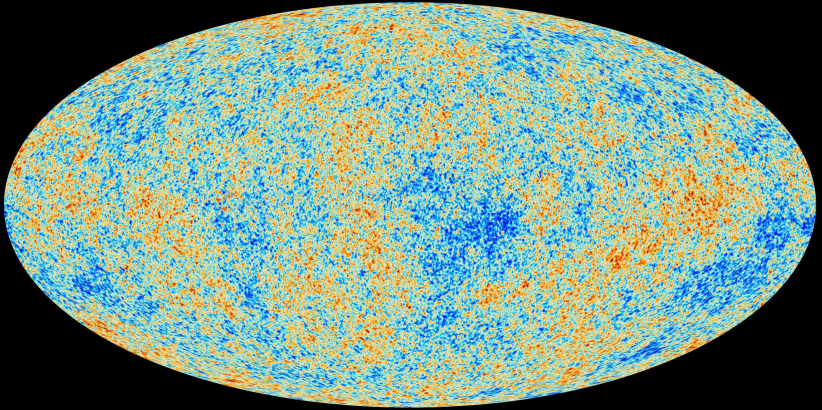
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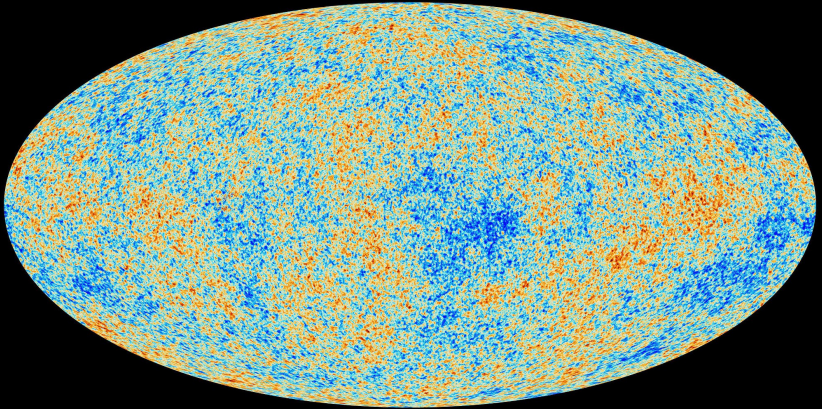
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- ▶ Red $\Rightarrow$ higher density

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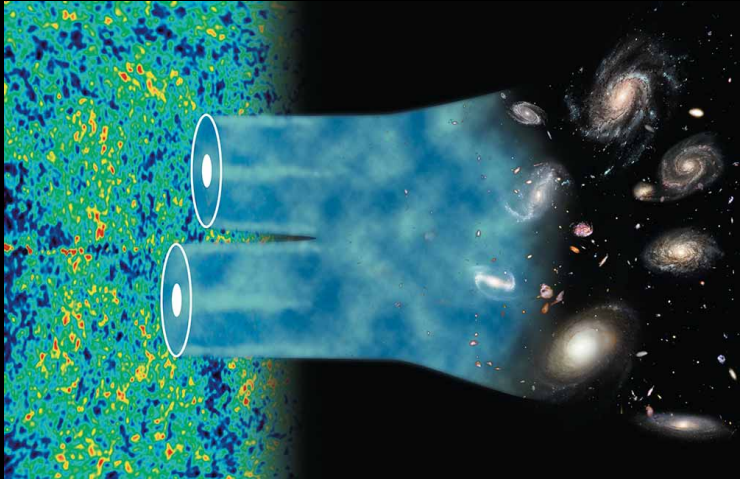
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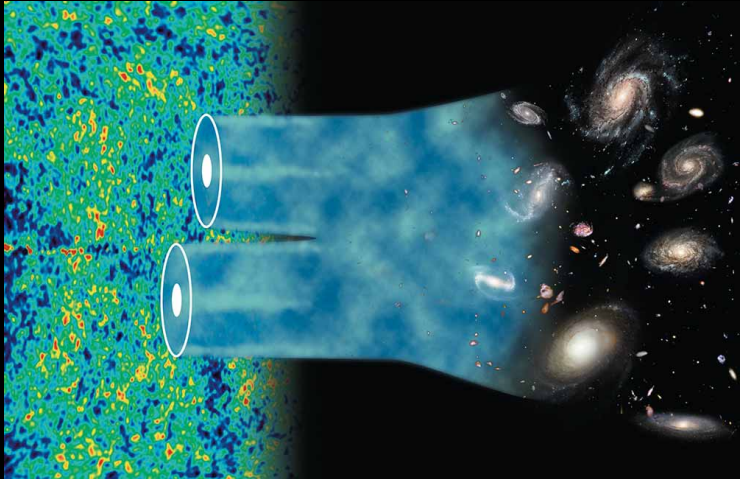
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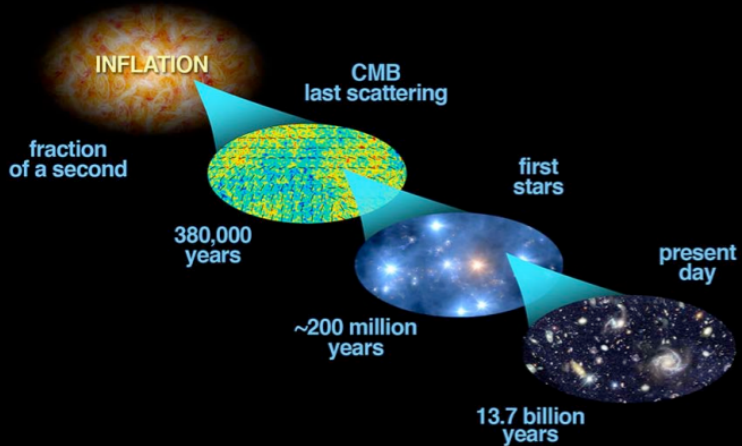
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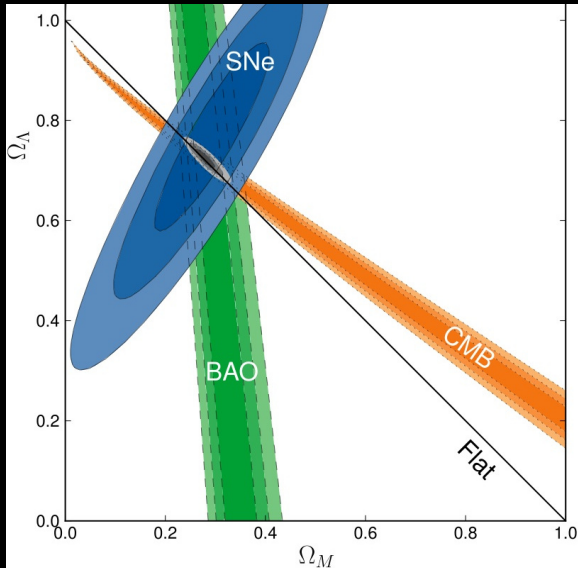
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Science from the CMB

The amount of dark matter and dark energy

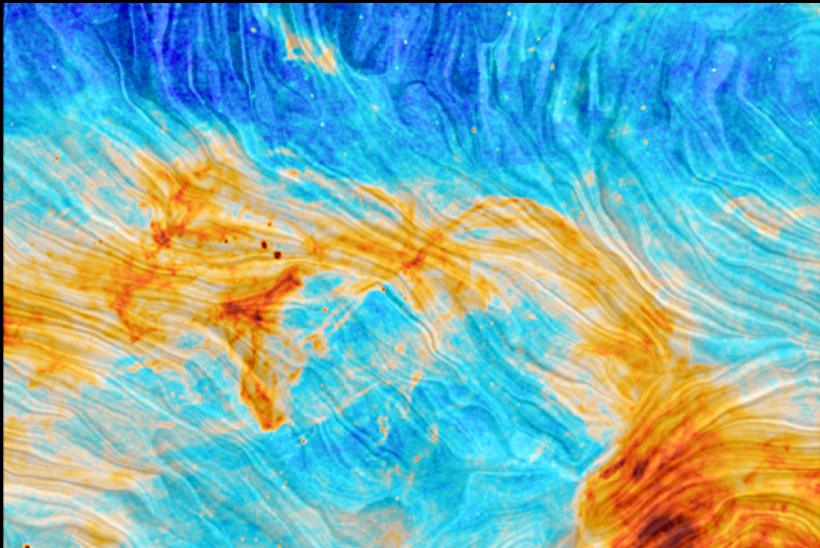


More science to come

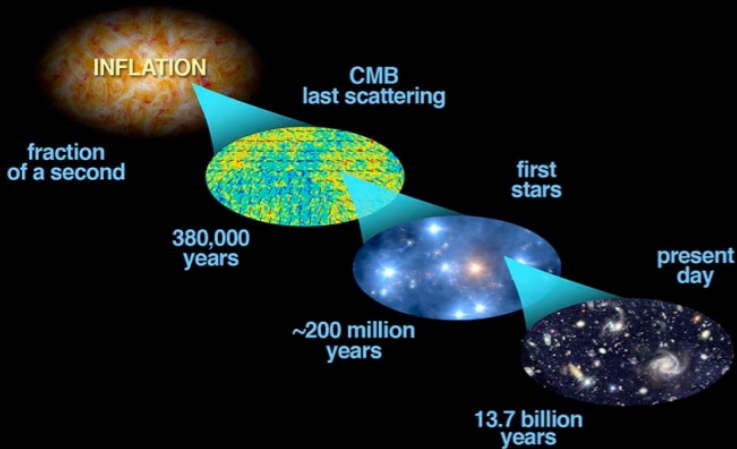
Polarisation

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Conclusions



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